4 The functions p and q are given by $p(x) = x^2 - 3x$ and $q(x) = 2x - 6$, $x \in \mathbb{R}$.
Find the two values of x for which $p(x) = q(x)$.

5 The functions f and g are given by $f(x) = 2x^3 + 30x$ and $g(x) = 17x^2$, $x \in \mathbb{R}$.
Find the three values of x for which $f(x) = g(x)$.

E **6** The function f is defined as $f(x) = x^2 - 2x + 2$, $x \in \mathbb{R}$.

a Write $f(x)$ in the form $(x + p)^2 + q$, where p and q are constants to be found. **(2 marks)**

b Hence, or otherwise, explain why $f(x) > 0$ for all values of x, and find the minimum value of $f(x)$. **(1 mark)**

7 Find all roots of the following functions:

a $f(x) = x^6 + 9x^3 + 8$

b $g(x) = x^4 - 12x^2 + 32$

c $h(x) = 27x^6 + 26x^3 - 1$

d $j(x) = 32x^{10} - 33x^5 + 1$

e $k(x) = x - 7\sqrt{x} + 10$

f $m(x) = 2x^{\frac{2}{3}} + 2x^{\frac{1}{3}} - 12$

Hint The function in part **b** has four roots.

E/P **8** The function f is defined as $f(x) = 3^{2x} - 28(3^x) + 27$, $x \in \mathbb{R}$.

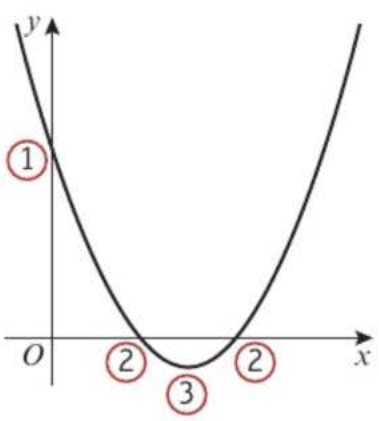
a Write $f(x)$ in the form $(3^x - a)(3^x - b)$, where a and b are real constants. **(2 marks)**

b Hence find the two roots of $f(x)$. **(2 marks)**

Problem-solving
Consider $f(x)$ as a function of a function.

2.4 Quadratic graphs

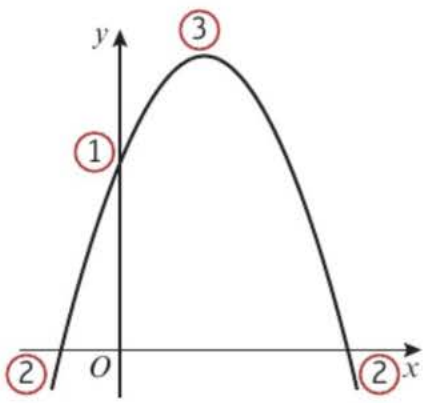
When $f(x) = ax^2 + bx + c$, the graph of $y = f(x)$ has a curved shape called a parabola.
You can sketch a quadratic graph by identifying key features.
The coefficient of x^2 determines the overall shape of the graph.
When a is positive the parabola will have this shape: ∪
When a is negative the parabola will have this shape: ∩



(1) The graph crosses the y-axis when $x = 0$. The y-coordinate is equal to c.

(2) The graph crosses the x-axis when $y = 0$. The x-coordinates are roots of the function $f(x)$.

(3) Quadratic graphs have one turning point. This can be a minimum or a maximum. Since a parabola is symmetrical, the turning point and line of symmetry are half-way between the two roots.



■ **You can find the coordinates of the turning point of a quadratic graph by completing the square. If $f(x) = a(x + p)^2 + q$, the graph of $y = f(x)$ has a turning point at $(-p, q)$.**

Links The graph of $y = a(x + p)^2 + q$ is a translation of the graph of $y = ax^2$ by $\begin{pmatrix} -p \\ q \end{pmatrix}$. → Section 4.5

Example 11

Sketch the graph of $y = x^2 - 5x + 4$, and find the coordinates of its turning point.

As $a = 1$ is positive, the graph has a $\vee$ shape and a minimum point.

When $x = 0$, $y = 4$, so the graph crosses the y-axis at $(0, 4)$.

When $y = 0$,

$$x^2 - 5x + 4 = 0$$

$$(x - 1)(x - 4) = 0$$

$x = 1$ or $x = 4$, so the graph crosses the x-axis at $(1, 0)$ and $(4, 0)$.

Completing the square:

$$x^2 - 5x + 4 = \left(x - \tfrac{5}{2}\right)^2 - \tfrac{25}{4} + 4$$

$$= \left(x - \tfrac{5}{2}\right)^2 - \tfrac{9}{4}$$

So the minimum point has coordinates $\left(\tfrac{5}{2}, -\tfrac{9}{4}\right)$.

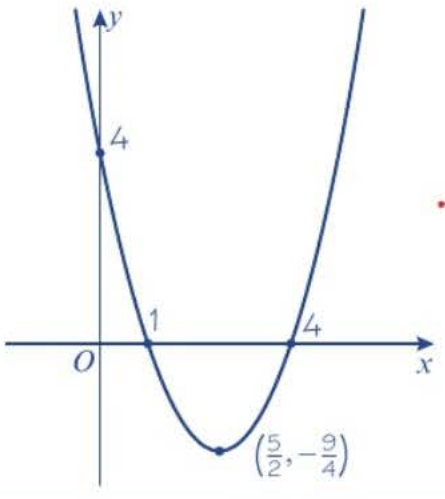
Alternatively, the minimum occurs when x is half-way between 1 and 4,

so $x = \dfrac{1 + 4}{2} = \tfrac{5}{2}$

$$y = \left(\tfrac{5}{2}\right)^2 - 5 \times \left(\tfrac{5}{2}\right) + 4 = -\tfrac{9}{4}$$

so the minimum has coordinates $\left(\tfrac{5}{2}, -\tfrac{9}{4}\right)$.

The sketch of the graph is:



Use the coefficient of x^2 to determine the general shape of the graph.

This example factorises, but you may need to use the quadratic formula or complete the square.

Complete the square to find the coordinates of the turning point.

Watch out If you use symmetry to find the x-coordinate of the minimum point, you need to substitute this value into the equation to find the y-coordinate of the minimum point.

You could use a graphic calculator or substitute some values to check your sketch.

When $x = 5$, $y = 5^2 - 5 \times 5 + 4 = 4$.

Online Explore how the graph of $y = (x + p)^2 + q$ changes as the values of p and q change using GeoGebra.

Example 12

Sketch the graph of $y = 4x - 2x^2 - 3$. Find the coordinates of its turning point and write down the equation of its line of symmetry.

As $a = -2$ is negative, the graph has a $\cap$ shape and a maximum point.

When $x = 0$, $y = -3$, so the graph crosses the y-axis at $(0, -3)$.

When $y = 0$,

$-2x^2 + 4x - 3 = 0$

Using the quadratic formula,

$$x = \frac{-4 \pm \sqrt{4^2 - 4(-2)(-3)}}{2 \times (-2)}$$

$$x = \frac{-4 \pm \sqrt{-8}}{-4}$$

There are no real solutions, so the graph does not cross the x-axis.

Completing the square:

$-2x^2 + 4x - 3$

$= -2(x^2 - 2x) - 3$

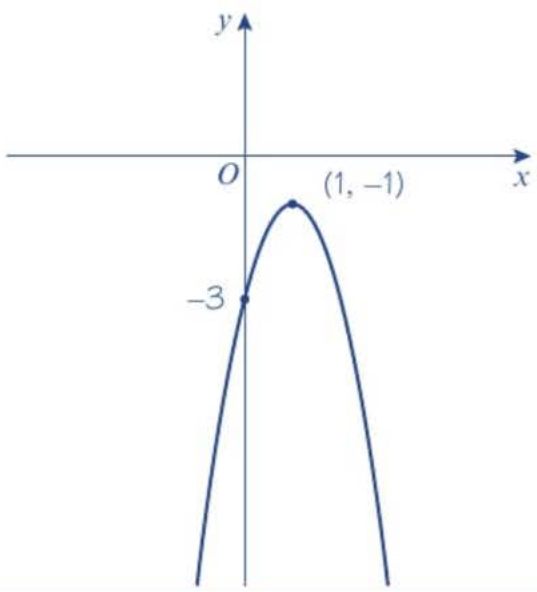
$= -2((x - 1)^2 - 1) - 3$

$= -2(x - 1)^2 + 2 - 3$

$= -2(x - 1)^2 - 1$

So the maximum point has coordinates $(1, -1)$.

The line of symmetry is vertical and goes through the maximum point. It has the equation $x = 1$.



It's easier to see that $a < 0$ if you write the equation in the form $y = -2x^2 + 4x - 3$.

$a = -2$, $b = 4$ and $c = -3$

You would need to square root a negative number to evaluate this expression. Therefore this equation has no real solutions.

The coefficient of x^2 is -2 so take out a factor of -2

Watch out A sketch graph does not need to be plotted exactly or drawn to scale. However you should:

- draw a smooth curve by hand
- identify any relevant key points (such as intercepts and turning points)
- label your axes.

Exercise 2F

1 Sketch the graphs of the following equations. For each graph, show the coordinates of the point(s) where the graph crosses the coordinate axes, and write down the coordinate of the turning point and the equation of the line of symmetry.

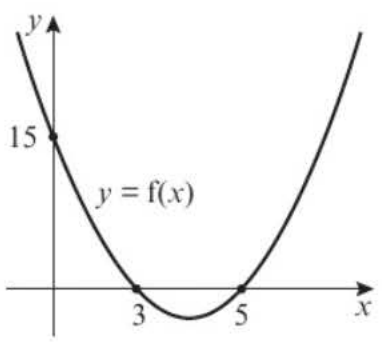
a $y = x^2 - 6x + 8$ **b** $y = x^2 + 2x - 15$ **c** $y = 25 - x^2$ **d** $y = x^2 + 3x + 2$

e $y = -x^2 + 6x + 7$ **f** $y = 2x^2 + 4x + 10$ **g** $y = 2x^2 + 7x - 15$ **h** $y = 6x^2 - 19x + 10$

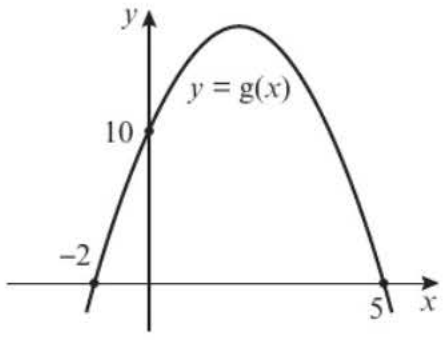
i $y = 4 - 7x - 2x^2$ **j** $y = 0.5x^2 + 0.2x + 0.02$

(P) **2** These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$.
Find the values of a, b and c for each function.

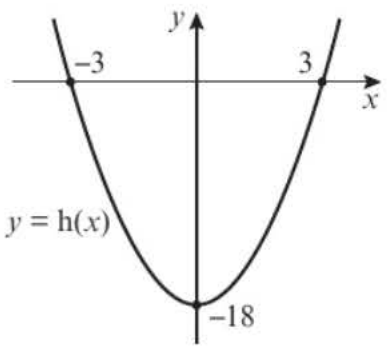
a



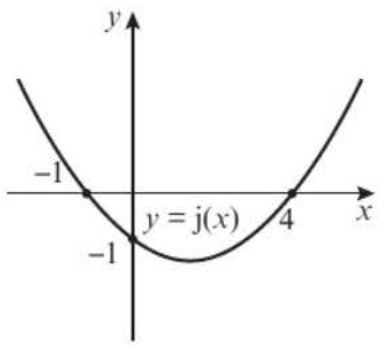
b



c



d



Problem-solving

Check your answers by substituting values into the function. In part **c** the graph passes through (0, –18), so h(0) should be –18.

(E/P) **3** The graph of $y = ax^2 + bx + c$ has a minimum at (5, –3) and passes through (4, 0).
Find the values of a, b and c. **(3 marks)**

2.5 The discriminant

If you square any real number, the result is greater than or equal to 0. This means that if y is negative, $\sqrt{y}$ cannot be a real number. Look at the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If the value under the square root sign is negative, x cannot be a real number and there are no real solutions. If the value under the square root is equal to 0, both solutions will be the same.

- **For the quadratic function $f(x) = ax^2 + bx + c$, the expression $b^2 - 4ac$ is called the discriminant. The value of the discriminant shows how many roots $f(x)$ has:**
  - **If $b^2 - 4ac > 0$ then $f(x)$ has two distinct real roots.**
  - **If $b^2 - 4ac = 0$ then $f(x)$ has one repeated root.**
  - **If $b^2 - 4ac < 0$ then $f(x)$ has no real roots.**